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Machine learning-enhanced IoT network security: a Black Hole Algorithm-based feature selection approach for intrusion detection

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ABSTRACT

The proliferation of Internet of Things (IoT) devices has substantially increased the vulnerability to cyber threats, worsened by limited computational capabilities and weak security mechanisms. To address these challenges, this study proposes a machine learning-based Intrusion Detection System (IDS) that leverages an optimized feature selection mechanism using the Black Hole Algorithm (BHA). The approach introduces a novel star encoding scheme and a hybrid fitness function to enhance solution convergence and classification performance. The methodology is evaluated using the AWID3 dataset, wherein feature preprocessing and selection are followed by classification using the XGBoost model. Experimental results demonstrate significant improvements over existing techniques, achieving an accuracy of 99.63%, precision of 99.57%, recall of 99.73%, F1-score of 99.56%, a false positive rate of 0.49%, and significant improvements in FNR, TPR, and TNR. These results outperform benchmark IDS models, including HHGS-ROA, GA-GWO, and BGWO. Furthermore, statistical analysis of the simulation results confirms the findings and demonstrates significant improvements in performance. This work establishes an efficient and scalable IDS framework for IoT wireless networks by combining metaheuristic optimization with machine learning classification, thereby offering a practical solution for real-time threat detection in resource-constrained environments.

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IoT security; Black Hole algorithm; intrusion detection; feature selection; machine learning

1. Introduction

The intrinsic characteristics of Internet of Things (IoT) networks, including resource limitations, scalability, and heterogeneity, pose a serious challenge to security and data privacy [1,2]. The fact that IoT devices frequently gather and send sensitive data makes them appealing targets for cyberattacks, which further exacerbates these difficulties. In this context, cybersecurity professionals have made numerous attempts to strengthen IoT security, such as encrypting data sent over networks, updating firmware often, and employing security keys and strong passwords [3,4]. However, due to their diversity, IoT devices remain vulnerable to a wide range of threats despite the aforementioned remedies. The industry is currently focusing on strategies to reduce the harm brought about by intrusions into IoT devices. IDSs, which are a key part of cybersecurity, are now a crucial tool for detecting network attacks. Thus, IDSs have gradually evolved into an essential network security technology by actively monitoring network and system activities in real-time and spotting possible malicious behaviors by examining patterns, indicators, and anomalies in network traffic data [5,6].

An IDS continuously checks the incoming data traffic generated by different devices and analyzes the data to identify potential occurrences of cyber-threats [7,8]. In other words, an IDS serves as an alert or network monitor. By sounding a warning before the intruder launches an attack, it keeps the system from being destroyed [9]. Typically, IDS architectures consist of two stages: a preprocessing phase, which includes data cleaning, feature extraction, and feature selection, and a classification phase, which distinguishes between benign and malicious behavior [10]. Herein, Feature selection (FS) is crucial for enhancing classification accuracy by reducing dimensionality and eliminating redundant or irrelevant attributes. Feature selection (FS) methodologies aim to minimize redundancy while maximizing relevance, which enhances detection

accuracy and improves model interpretability. This also reduces computational overhead, which is especially important in big data scenarios where real-time intrusion detection is essential.

The classification process involves building a model (classifier) from labeled training datasets and using it to predict categories for new input traffic [11]. Many IDSs use either a single-class or multi-class classification framework. While single-class classification assumes all inputs belong to one normal class, multi-class classifiers can distinguish among various attack types and normal traffic, typically achieving higher detection accuracy [12]. However, as feature dimensionality increases, existing IDSs often experience degraded performance with respect to both accuracy and processing efficiency [10,13].

To address these limitations, the adoption of machine learning (ML) approaches has gained momentum due to their ability to automatically analyze vast, continuous data streams generated by IoT devices. ML techniques show promise in identifying irregular routing behaviors and attack patterns, but often suffer performance degradation as data dimensionality increases [14–16]. In resource-constrained environments, this necessitates the use of intelligent FS strategies that can guarantee both optimal model accuracy and runtime efficiency. Within this context, bio-inspired metaheuristic optimization algorithms have emerged as viable solutions for high-efficiency FS. Approaches such as Particle Swarm Optimization (PSO), Genetic Algorithms (GA), and other evolutionary strategies have demonstrated encouraging results in identifying relevant feature subsets [17]. Particularly, the Black hole (BH) algorithm [18] offers a remarkably simple structure with only one controlling parameter (the radius of the black hole) and demonstrates faster convergence rates compared to other metaheuristics like PSO and GA, enabling efficient exploration of complex solution spaces typical in IoT security scenarios [19].

Keeping in view the above discussion, we propose a novel intrusion detection system (IDS) for IoT wireless networks, leveraging an advanced feature selection technique that combines the Black Hole Algorithm (BHA) with a distinctive star encoding mechanism and an efficient multi-objective fitness function. The scheme effectively identifies the most relevant features from high-dimensional network datasets, substantially reducing computational complexity and improving detection accuracy. By integrating the optimized feature subset with the XGBoost classifier, the system achieves superior performance in distinguishing between benign and malicious traffic. The novel star encoding for feature representation and the fitness function facilitate efficient and interpretable feature selection, which is particularly important for addressing the unique security challenges in IoT environments. The approach offers a scalable, real-time, and resource-efficient IDS framework, making it highly suitable for deployment in complex and dynamic IoT networks. The major contributions of the proposed scheme are summarised as follows:

- We propose a novel feature selection approach for IDSs based on the Black Hole Algorithm (BHA), introducing a star encoding mechanism and a novel fitness function to optimize the search for relevant features in IoT network data.
- We develop an efficient machine learning framework that integrates the proposed BHA-based feature selection with an XGBoost classifier to enhance intrusion detection accuracy and computational performance in IoT wireless network environments.
- We perform comprehensive experimental evaluations of the AWID3 dataset, demonstrating the superior performance of the proposed framework, achieving 99.63% precision, 99.57% precision, 99.73% recall, 99.56% F1-score, a false positive rate of 0.49%, and significant improvements in FNR, TPR, and TNR compared to several state-of-the-art IDS models. The statistical analysis further confirms the performance gains of the proposed method in terms of the simulation results.
- We provide in-depth analysis of dimensionality and computational efficiency, showing that the proposed method significantly reduces the number of features, making it suitable for real-time and resource-constrained IoT applications.

2. Related works

In [13], the authors proposed a hybrid Hunger Games Search and Remora Optimization Algorithm (HHGS-ROA) to solve the security issues related to IDS in IoT. The proposed method for an Internet of Things IDS makes use of the actual Aegean Wi-Fi Intrusion Dataset (AWID). Initially, the preprocessing step uses the AWID raw recorded traffic to speed up feature selection. The method used a hybrid HGS-ROA model to

extract the AWID dataset's features. In order to distinguish between malicious traffic and regular network traffic, the SVM classifier is finally used. Thus, the suggested method combines an improved detection rate with a low false-positive rate (FPR). The approach is centered on hybrid metaheuristics utilizing Support Vector Machines (SVM); however, it suffers from limited interpretability and imposes substantial computational overhead during the feature selection process. These limitations significantly restrict its suitability for real-time applications in resource-constrained IoT environments.

In [20], the authors presented an IDS system based on GWO-PSO, which was intended to increase security by detecting various IoT network assaults. The pertinent network features were extracted using GWO-PSO from the data input. Ultimately, the random forest (RF) method was used to complete the detection process. To improve the detection ratio, RF examined the features that have been extracted. The CICIDS-2017, NSL-KDD, and KDDCup99 datasets are the three distinct datasets used in this procedure to verify the GWO-PSO process's performance. Additionally, it overcomes the problem of data imbalance to produce superior results. Although this approach has demonstrated improved detection performance using Random Forest (RF) classifiers, it primarily relies on general-purpose datasets such as CICIDS-2017 and NSL-KDD, without evaluation on IoT-specific datasets like AWID3. This limitation constrains its domain relevance and applicability to IoT environments.

An adversarial decomposition and neighborhood evolution model based on the dynamic multi-objective particle swarm optimization technique was presented by [21]. A dynamic reaction mechanism that included global optimal information exploration and archive set prediction was demonstrated. This was carried out in order to facilitate environmental adaptation. The approach employs a deep multi-objective framework but lacks lightweight design principles, which are critical for IoT environments. A deep learning method for identifying intrusions into IoT wireless networks was proposed in [22]. Using a novel approach to identifying intrusions into an IoT wireless network, deep learning-based approaches are proposed in this procedure to determine whether the traffic flow is regular or if an attack is occurring. Initially, characteristics are extracted from packet-level field data to create the traffic flow in the dataset. The traffic flow classification in IoT networks is either normal, or the assault is identified by the classification of distributed denial of service, denial of service, information theft attack, and reconnaissance, which yields a high classification accuracy. The scheme achieves high accuracy but is computationally intensive, making it impractical for real-time or low-power deployments common in wireless IoT networks.

Using hybrid metaheuristic algorithms, the authors in [23] proposed an IoT IDS for an efficient feature selection model. In order to improve the Gorilla Troops Optimizer (GTO), which is based on the Bird Swarms (BSA) algorithm, this work employs a novel feature selection technique. It pinpoints the viable areas with the best answers to improve GTO-BSA performance exploitation. Higher quality findings are obtained, increasing convergence with four IoT-IDS datasets, including NSL-KDD, CICIDS-2017, UNSW-NB15, and BoT-IoT. A variety of performance measures were used to evaluate GTO-BSA's performance. The outcomes were contrasted with those of the original GTO, BSA, and a number of other innovative methods found in the literature. The tests showed that GTO-BSA produced higher-quality solutions and had a higher rate of convergence. Even though it improves convergence speed and feature subset optimization, it involves high algorithmic complexity and does not integrate a classification framework optimized for the reduced feature set.

A lightweight IDS is presented by the authors in [24] for Internet of Things networks, which produce a significant amount of false alarms. Usability issues for real-world evidence include traffic restriction policies that track suspicious network traffic, such as a high TCP connection rate. Several other works [25–27] exist on IDS for IoTs. However, these approaches don't produce improved IDS results because of certain limitations. Considering the shortcomings of the above works, we proposed a novel BHA-based IDS in the following. The approach attempts to minimize false alarms in IoT, yet fails to optimize feature relevance effectively, resulting in compromised detection accuracy when facing sophisticated or zero-day attacks.

The existing body of research on intrusion detection systems (IDS) for IoT security exhibits several notable limitations, including high computational demands, inefficient or non-optimized feature selection mechanisms, dependence on outdated or non-IoT-specific datasets, and insufficient support for real-time performance in resource-constrained environments. While deep learning-based methods typically achieve high detection accuracy, they are often hindered by significant computational overhead, whereas hybrid metaheuristics may lack interpretability and require extensive parameter tuning. Additionally, few studies adequately address feature interpretability or enable dynamic adaptation to evolving intrusion patterns. To address these challenges, the proposed work presents a lightweight, interpretable, and computationally

efficient IDS framework that employs a novel Black Hole Algorithm-based feature selection technique, augmented by star encoding and a customized fitness function. By integrating this feature selection with an XGBoost classifier and evaluating the system on the IoT-specific AWID3 dataset, the proposed approach effectively overcomes the limitations of existing solutions, establishing a new standard for accurate, real-time, and scalable intrusion detection in IoT environments.

3. Proposed methodology

In this section, we propose a novel machine-learning architecture of intrusion detection for WSNs that utilizes the BH algorithm as shown in [Figure 1](#). The aim is to improve the efficiency of the system while keeping the utilization of resources limited. The proposed IDS is designed as a multi-stage architecture comprising four primary modules: (i) Data Preprocessing, (ii) Feature Selection, (iii) Model Training and Classification, and (iv) Intrusion Detection Output. Initially, raw traffic data from the AWID3 dataset is cleaned and normalized to remove missing, symbolic, or non-numeric values. This processed dataset is then passed into the Feature Selection module, where a novel Black Hole Algorithm (BHA)-based approach, supported by the star encoding representation and a custom fitness function, identifies a minimal yet highly relevant subset of features for classification. The selected features are then fed into the XGBoost classifier, which is trained to distinguish between benign and malicious traffic classes using gradient-boosted decision trees. Finally, the Intrusion Detection Output layer generates predictions and calculates evaluation metrics such as accuracy, precision, recall, FPR, and F1-score. Together, these modules form an efficient, scalable IDS suitable for real-time threat detection in resource-constrained IoT environments. Each of the modules is discussed in detail in the following sections. However, we first start with the dataset description.

3.1. Dataset description

In the work, we used the Aegean Wi-Fi Intrusion Dataset (AWID3) [28], which is a comprehensive dataset designed to aid research in wireless network intrusion detection, particularly in Wi-Fi environments. It was developed to address the challenges of detecting intrusions and malicious activities in IEEE 802.11 networks. The dataset captures a wide range of activities, including normal traffic, as well as various attack types like de-authentication, disassociation, and man-in-the-middle attacks. It includes extensive network traffic traces collected under different conditions and scenarios, encompassing both benign and malicious behaviors. Each record in the dataset contains numerous features extracted from the 802.11 headers, including frame control fields, sequence numbers, and time intervals, which can be analyzed for anomaly detection and classification tasks.

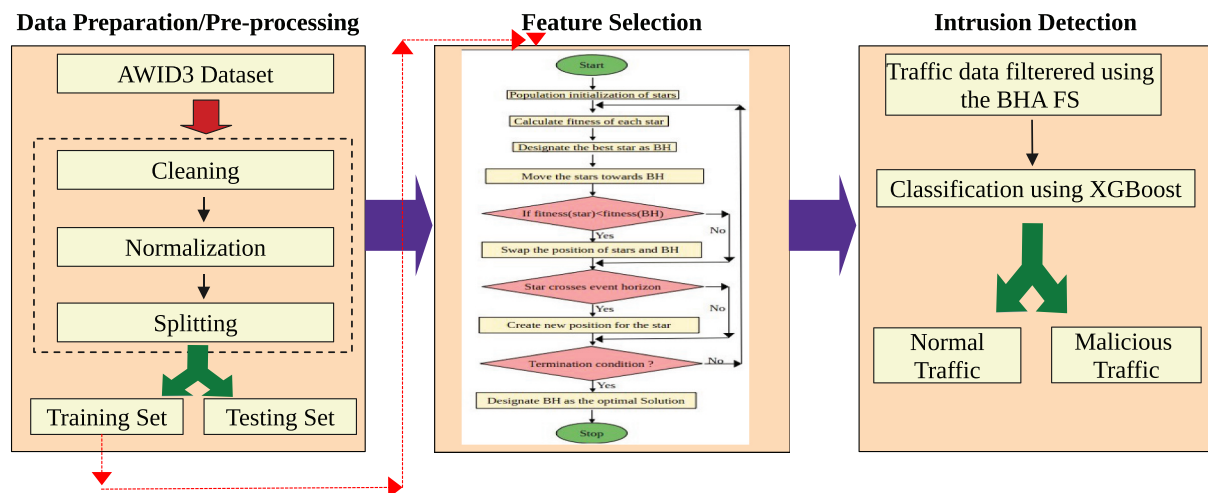


Figure 1. Working model of the proposed BHA-based IDS.

Actual Wi-Fi traffic is represented by the AWID3 dataset. Generally, the AWID3 dataset is used in the measurements of wireless network IDSs. It is structured into two subfolders for detailed analysis of attack scenarios. The dataset contains 13 target classes such as 'Death', 'Dias', '(Re)Assoc', etc. The dataset is sufficiently large, containing millions of records. AWID3 concentrates on Protected Management Frameworks (PMF) and WPA2-Enterprise. It covers 13 attacks, from high-level and more sophisticated ones like Kr00k, Krack, malware, botnets, and amplification to more traditional deauthentication. The attacks are diversified to the MAC layer and higher layers. It is available in pcap (raw data, complete set of all potential features) and CSV (254 features) formats. Data preparation is done using the proposed algorithm to enhance the speed of choosing helpful traffic. The chosen traffic is then fed into XGBoost to categorize the incursion in the IoT IDSs.

3.2. Data preparation

There are three stages in the data preparation stage: cleaning, normalization, and splitting, as shown in Figure 1. Herein, the raw AWID3 data is first cleaned to make the data more meaningful for processing. Next, normalization converts alpha-numeric values into suitable numeric values. Finally, the processed data is split into training and testing parts.

The first phase in data preparation is data cleaning, as shown in Figure 1. This step speeds up the process of feature extraction. Here, certain values of the dataset are replaced with some special character, such as a hyphen (used in the study), etc. In other words, this ensures that the values of some attributes (columns) in the dataset are not available. Next, the dataset contains notational or symbolic values for several attributes, such as receiver address, basic service set identifier (network identifier), frame type (e.g. management, control, data), etc., that are mapped to a suitable integer value. For example, frame type 'management': 0, 'control': 1, 'data': 2, etc.

The second phase is called the normalization phase. In this phase, the non-numeric values of attributes are transformed into integer values. Note that the dataset is characterized by a large number of outliers; therefore, we employ the min-max normalization, expressed as follows:

$$x_n = \frac{X - X_{min}}{X_{max} - X_{min}} \quad (1)$$

where x_n is the normalized value of an original value x in the dataset. After this phase is over, the dataset is processed through the splitting stage. Here, the dataset is randomly divided into the testing and training sets. After preprocessing the AWID3 dataset, a feature selection approach based on the Black Hole Optimization Algorithm is performed. Note that feature selection is crucial after preprocessing to reduce dimensionality, improve model performance, and prevent overfitting by removing irrelevant or redundant features. Further, achieving high accuracy and low computing complexity improves intrusion detection in IoT wireless networks. Keeping this in view, we use the BH algorithm to select features. BHA is suitable for feature selection in IDSs because it efficiently identifies optimal feature subsets from high-dimensional data, improving detection accuracy and reducing computational overhead. Its ability to handle nonlinearity and dynamic search spaces makes it ideal for the complex nature of intrusion patterns.

3.3. BHO-based feature selection

The Black Hole Algorithm (BHA) is a metaheuristic optimization technique inspired by black hole dynamics, where candidate solutions are attracted toward the best solution, simulating gravitational pull. It excels in solving complex optimization problems by balancing exploration and exploitation to avoid local optima. Similar to other population-based algorithms, the proposed Black Hole Algorithm (BH) begins with a randomly generated population of candidate solutions, referred to as stars, distributed across the search space of the problem or function. After initialization, the fitness/objective values of these candidates are evaluated, and the best solution is designated as the black hole, while the remaining candidates become regular stars. The black hole exerts an attractive pull, absorbing nearby stars into its vicinity. Mathematically, this is represented by the following equation:

$$X_i^{t+1} = X_i^t + rand() \times (X_B - X_i^t), i = 1, 2, \dots, n^p \quad (2)$$

where X_i^{t+1} is the positions of the i^{th} star in the t^{th} iteration, respectively. X_B represents the position of the BH, $rand()$ is a function that generates a random number in the interval $[0, 1]$, and n^p is the size of the population of the stars. A star might travel toward the black hole and end up at a place with a lower fitness value than the BH. The BH moves to the location of that star, in this case, and vice versa. The BH algorithm will then proceed with the new black hole, causing stars to start moving in its direction. It is also possible for stars traveling toward the black hole to reach the event horizon. Any star (candidate solution) that crosses the black hole's event horizon will be absorbed. Every time a candidate (star) dies, the black hole consumes it – a new search is started, with a new candidate solution (star) generated and distributed at random across the search space. This is done to keep the number of possible solutions constant. The next iteration starts after each star has been moved. Mathematically, the event horizon (r) radius is expressed as:

$$r = \frac{O_B^F}{\sum_{i=1}^{n^p} O_i^F} \quad (3)$$

where O_B^F is the fitness function value for the BH and O_i^F is the fitness function value of the i^{th} star. n^p is the size of the population or the number of stars. When the distance between a star and the BH (the best candidate) is less than r , the star collapses, and a new star is created and distributed randomly throughout the search space. This process is repeated a sufficient number of times until all stars are absorbed by the BH. In other words, the algorithm is repeated until an optimal solution is found. Considering the feature selection process, the process is repeated until an optimal set of features is selected. The working diagram of the BHA is represented in Figure 2.

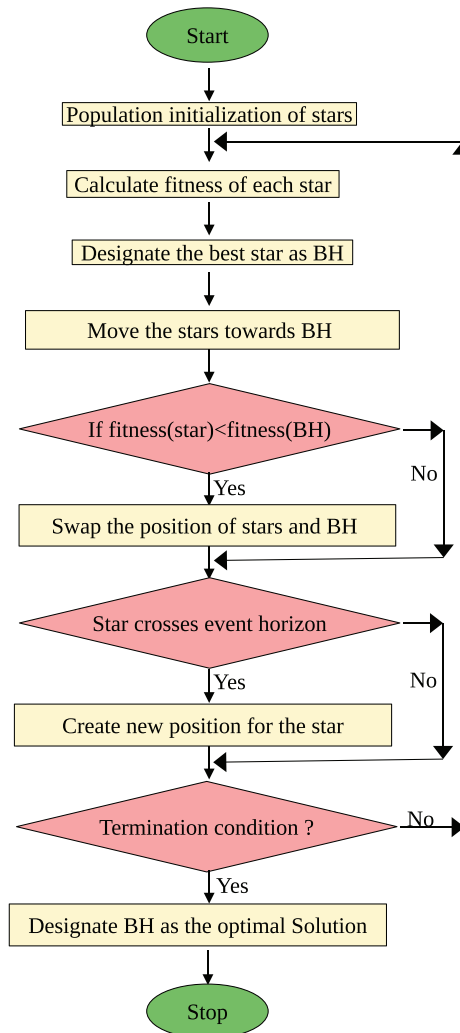


Figure 2. Working of the BH algorithm for optimal feature selection.

(1,.212)	(1,.723)	(0,0.3)	(0,0.199)	(1,.869)	(1,0.267)	(0,0.112)	(0,.921)	(1,.342)
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Figure 3. Typical encoding of a star in the BH algorithm for feature selection.

3.4. Star representation

A star represents a complete solution for feature selection. In other words, it provides the number of features selected and also the weights assigned to each feature. A star is represented as $[s_1, s_2, \dots, s_n^f]$. Each $s_i, 1 \leq i \leq n^f$ has two components: (1) $b_i, 1 \leq i \leq n^f$: A binary variable having value (0 or 1). It indicates whether the feature is selected or rejected. (2) $w_i, 1 \leq i \leq n^f$: A variable having value in the range[0,1]. It indicates how much weight is assigned to a feature. In other words, each candidate solution $\mathbf{x}_i = [b_i, w_i]$ comprises a binary selection indicator b_i and a continuous feature-importance weight w_i . During each iteration of the Black Hole Algorithm, the position update rule

$$\mathbf{x}_i^{t+1} = \mathbf{x}_i^t + \text{rand}() \times (\mathbf{x}_B^t - \mathbf{x}_i^t),$$

where $\text{rand}() \sim U(0, 1)$ and $\mathbf{x}_B^t$ denotes the current best solution, is applied. This update simultaneously adjusts both the binary mask b_i and the continuous weight w_i , guiding w_i progressively toward the optimal solution and ensuring that feature-importance weights converge alongside the selection mask. A sample particle representation is shown in Figure 3.

In the proposed work, we use a weighted feature encoding method for the star. The dimension of the particle is the number of features in the dataset after preprocessing. As shown in the figure, there are nine cells in the star. Thus, the dimension of the star is nine. Each cell has two values associated with it. The first value is a binary value in the interval [0,1]. If the feature is selected, then the value one is selected; otherwise, zero is selected. The second value in the cell refers to the weight of the feature selected. It signifies how important that feature is in deciding whether an intrusion has occurred or not. Note that some features that are not selected ($b_i = 0$) still are assigned weight (i.e. $w_i > 0$). This arises due to the random initialization of stars. However, it is trivial and done for the sake of uniformity of representation and simplicity of computations.

The BH algorithm consists of a group of candidate solutions (stars). Each star is initialized in a similar manner, as explained earlier. Thus, a population of stars is formed, representing possible solutions of the search space. Each star has an associated fitness value given by a fitness as explained in Section 3.5 The star that has the best fitness value is considered to be the black hole. Specifically, the star that has the lowest fitness function is called the black hole.

3.5. Fitness function

As described in Section 3.4, each star represents a solution to the optimal selection of features from the dataset to improve the accuracy of the classification of data to determine whether it is an intrusion or not. Thus, a fitness function is used to measure how good a solution is represented via a star. The fitness function used here is a combination of two different components. The first component is the classification accuracy (F_1), which accurately identifies attack and non-attack activities. Mathematically, it is represented as follows:

$$F_1 = \frac{T_P + T_N}{T_P + T_N + F_P + F_N} \quad (4)$$

where T_P represents the correctly labeled attack, T_N represents the correctly labeled normal case, the wrongly labeled attack case, and T_n represents the wrongly labeled normal case.

The second component of the fitness function is called the false alarm rate. It measures the instances of attacks wrongly identified. It is expressed as follows:

$$F_2 = \frac{F_P}{F_P + T_N} \quad (5)$$

where F_p and T_N have the same meaning as explained earlier. Note that the two objectives are conflicting in nature [29]. Therefore, the fitness function is a nonlinear combination of the two objectives. Thus, the fitness function is expressed as follows:

$$F = \text{Minimize } \alpha \times F_1 + \beta \times \frac{1}{F_2} \quad (6)$$

where $\alpha + \beta = 1$. The fitness function considered is a minimization function. In other words, the smaller the value of the fitness function for a star, the better the solution represented by the star. Empirically, the weighting coefficients were set to $\alpha = 0.7$ and $\beta = 0.3$ in accordance with precedent multi-objective IDS studies that prioritize detection accuracy over false-alarm reduction, demonstrating that this configuration yields favorable Pareto-optimal trade-offs in wireless intrusion detection contexts. A subsequent sensitivity analysis confirmed that varying α within the range 0.6–0.8 consistently maximizes the F_1 -score without incurring a disproportionate increase in the false positive rate, thus aligning with established best practices in multi-objective feature-selection literature [30,31]. The BH algorithm, as applied to the IDS, is detailed in the following

Algorithm 1: Proposed BHA-Based Intrusion Detection Framework

Input: Raw IoT Network Dataset D
Output: Intrusion Detection Labels (Normal / Malicious)

- 1 **Stage 1: Data Preprocessing**
- 2 Clean D by replacing symbolic/missing values with numeric mappings ;
- 3 Normalize features using min-max scaling ;
- 4 Split data into training set D_{train} and test set D_{test} ;
- 5 **Stage 2: Feature Selection using BHA**
- 6 Initialize BH population: generate n_p stars with random feature subsets and weights ;
- 7 **foreach** star s_i *in* population **do**
- 8 Calculate fitness F_i using Equation 6:
- 9 $F_i = \alpha \cdot F_1 + \beta \cdot \frac{1}{FPR}$;
- 10 Identify best star s_b with minimum F_b (Black Hole) ;
- 11 **while** *termination condition not met* **do**
- 12 **foreach** star $s_i \neq s_b$ **do**
- 13 Update position using Equation 2:
- 14 $s_i^{t+1} = s_i^t + \text{rand}() \cdot (s_b - s_i^t)$;
- 15 Recalculate fitness F_i ;
- 16 **if** $F_i < F_b$ **then**
- 17 Swap $s_i \leftrightarrow s_b$;
- 18 Compute event horizon radius r using Equation 3 ;
- 19 **if** $\text{distance}(s_i, s_b) < r$ **then**
- 20 Regenerate s_i randomly in search space ;
- 21 Select optimal feature subset F^* from final Black Hole s_b ;
- 22 **Stage 3: Classification using XGBoost**
- 23 Train XGBoost classifier on F^* using D_{train} ;
- 24 Predict on D_{test} and output class labels ;

Algorithm 1 outlines a structured, three-phase workflow of the proposed Black Hole Algorithm (BHA)-based intrusion detection system (IDS) designed for IoT networks. In the data preprocessing phase, raw network traffic from the AWID3 dataset undergoes cleaning to handle symbolic and missing values, normalization using min-max scaling, and splitting into training and testing sets. In the feature selection phase, a population of candidate solutions (stars), each encoded with a binary feature selection mask and associated weights, is initialized. Each candidate is evaluated using a custom fitness function combining the F1-score and inverse false positive rate (FPR). The best candidate is designated as the black hole, and other stars are iteratively moved towards it based on a gravitational model. Stars crossing the event horizon are reinitialized, and this evolutionary process continues until convergence to an optimal feature subset. Finally, in the classification phase, the selected features are used to train an XGBoost classifier, which then predicts whether the network traffic instances are benign or malicious. This approach ensures improved detection accuracy, reduced dimensionality, and computational efficiency, making it suitable for real-time IoT security applications.

3.6. Classification using XGBoost

XGBoost is a scalable, high-performance machine learning library for gradient boosting, designed to optimize speed and accuracy, making it ideal for structured data tasks such as classification, regression, and ranking. XGBoost is suitable for intrusion detection due to its ability to handle high-dimensional data, making it ideal for datasets with numerous features after selection. Its robust regularization prevents overfitting, ensuring reliable performance on complex and imbalanced datasets. Moreover, its gradient-boosting framework efficiently captures nonlinear relationships, making it adept at identifying subtle patterns indicative of intrusions.

Feature selection ensures that only the most relevant features are used for intrusion detection, reducing noise and improving model performance. Assume, we have a dataset D with n^s samples and n^f features, denoted as $D = \{(x_1, y_1), (x_2, y_2), \dots, (x_{n^s}, y_{n^s})\}$, where $x_i \in \mathbb{R}^{n^f}$ is the feature vector for the i -th sample, and $y_i \in \{0, 1\}$ indicates whether the sample is normal ($y_i = 0$) or an intrusion ($y_i = 1$). After feature selection using the proposed method, we select a subset of features $\{f_1, f_2, \dots, f_k\}$ where $k \leq n^f$. The selected features $x_{i'} \in \mathbb{R}^k$ are used to train the XGBoost model.

XGBoost constructs an ensemble of decision trees to predict whether a sample is normal or an intrusion. It optimizes the following objective function over T iterations:

$$\mathcal{L}(\theta) = \sum_{i=1}^{n^s} l(y_i, \hat{y}_i) + \sum_{t=1}^T \Omega(f_t), \quad (7)$$

where $l(y_i, \hat{y}_i)$ is the loss function measuring the error between the true label y_i and the predicted probability $\hat{y}_i$. The term $\Omega(f_t)$ regularizes the complexity of the t -th tree f_t , defined as:

$$\Omega(f_t) = \gamma T + \frac{\lambda}{2} \sum_{j=1}^T w_j^2, \quad (8)$$

where T is the number of leaves, w_j is the weight of the j -th leaf, γ controls the regularization on the number of leaves, and λ penalizes large leaf weights. The model uses a gradient-boosting framework to iteratively minimize $\mathcal{L}(\theta)$.

The trained XGBoost model predicts the probability of an intrusion $\hat{y}_i = P(y_i = 1 | x_{i'})$ using the decision trees:

$$\hat{y}_i = \sigma \left(\sum_{t=1}^T f_t(x_{i'}) \right), \quad (9)$$

where $\sigma(z) = \frac{1}{1+e^{-z}}$ is the sigmoid function ensuring the output is a probability. A threshold τ (e.g. $\tau = 0.5$) is applied to classify the sample:

$$\text{Prediction} = \begin{cases} 1 & \text{if } \hat{y}_i \geq \tau, \\ 0 & \text{if } \hat{y}_i < \tau. \end{cases} \quad (10)$$

Features selected during preprocessing improve intrusion detection by reducing model complexity and enhancing interpretability, enabling XGBoost to focus on the most relevant dimensions of the data. This results in efficient and accurate intrusion detection.

Note that the reason behind choosing XGBoost over lighter classifiers is its superior ability to handle high-dimensional, imbalanced IoT traffic data through built-in regularization, parallel tree-construction, and out-of-core computing capabilities shown to reduce false alarms and improve classification robustness in IoT environments. While lightweight models (e.g. decision trees, Naïve Bayes) offer faster inference, they lack XGBoost's gradient-boosting optimization and regularization mechanisms, which are critical for achieving sub-1% false positive rates and maintaining high detection accuracy on complex and noisy IoT datasets.

4. Simulations

We compare the suggested work with the existing approaches using simulation. Specifically, Python was used to run the simulations on a Windows 11 computer with an i7 processor running at 2.10 GHz and 16 GB of RAM. By identifying intrusion, the proposed method aims to enhance the effectiveness of creating IoT systems. The dataset is divided into training set ($\alpha_1\%$) and testing set ($\alpha_2\%$) where $\alpha_1 = 70$ and $\alpha_2 = 30$. The simulation results are analyzed to classify the traffic as either normal or malicious. Table 1 details the parameters used in the simulation.

The proposed scheme is compared with competing schemes [13,32,33]. Specifically, the schemes are Binary Gray Wolf Optimization (BGWO) [34], Genetic Algorithm (FWP-SVM-GA) [35], HGS algorithms [36], hybrid Hunger Games Search and Remora Optimization Algorithm (HHGS-ROA) [13], and Genetic Algorithm-Gray Wolf Optimization (GAGWO) [37]. The reason behind the selection of comparative algorithms, including HHGS-ROA, GA-GWO, BGWO, FWP-SVM-GA, HGS, and ROA, is that they are state-of-the-art metaheuristic and hybrid approaches commonly applied in recent IoT intrusion detection system research, ensuring a rigorous and relevant benchmark for the proposed method. BGWO is a benchmark metaheuristic tailored for feature selection in IDS tasks that simulates social hierarchy and hunted prey search strategies. FWP-SVM-GA Combines Genetic Algorithms with SVM for classification tasks, a representative hybrid from traditional evolutionary ML models. GAGWO is an advanced IDS integrating Genetic Algorithm and Grey Wolf Optimizer, often studied for balancing global/local search in large feature spaces. HGS and ROA are utilized for nonlinear optimization in intrusion contexts; selected due to their capability to adaptively optimize IDS rules and threshold parameters. HHGS-ROA is a recent hybrid strategy combining Hunger Games Search with Remora Optimization, tailored for high-dimensional IoT network traffic.

The effectiveness of the proposed approach is evaluated by assessing its precision, accuracy, F1-score, false-positive rate, and recall for intrusion detection in IoT-based WSNs. A detailed description of the dataset is given in Section 3.1

Figure 4 shows the confusion matrix for the proposed scheduling algorithm. The matrix displays four concepts, true positive(T_P), true negative(T_N), false positive(F_P), and false negative(F_N). T_P is the number of times the normal traffic is correctly predicted as normal traffic. T_N is the number of times the malicious traffic(intrusion) is correctly predicted as malicious traffic. F_P is the number of times the malicious traffic is predicted as normal traffic. F_N is the number of times the normal traffic is predicted as malicious traffic.

Table 1. Simulation parameters and configuration details.

Category	Parameter	Description/Value
IoT Dataset (AWID3)	Dataset type	IEEE 802.11 wireless traffic (AWID3)
	Input format	CSV, 254 features per sample
	Number of classes	13 attack types + benign traffic
	Preprocessing	Symbolic-to-numeric mapping, min-max normalization, train-test split (70%-30%)
Feature Selection (BHA)	Algorithm used	Black Hole Algorithm (BHA)
	Population size	30 stars
	Iterations	100
	Encoding scheme	Star representation: binary (feature selected or not) + weight [0,1]
	Fitness function	$F = \alpha \cdot F1 + \beta \cdot \frac{1}{FPR}$, where $\alpha + \beta = 1$, $\alpha = 0.7$, $\beta = 0.3$
	Update mechanism	$x_i^{t+1} = x_i^t + rand() \cdot (x_{BH} - x_i^t)$
	Event horizon radius	$r = \frac{OF_{BH}}{\sum OF_i}$
	Termination condition	Maximum iterations or fitness value convergence
XGBoost Classifier Settings	Booster type	gbtree
	Number of estimators	100
	Max tree depth	6
	Learning rate (η)	0.1
	Subsample ratio	0.8
	Column sample by tree	0.8
	L2 regularization (λ)	1
	Min child weight	1
Execution Environment	Objective function	Binary logistic (malicious/normal prediction)
	Programming language	Python (v3.8+)
	Simulator platform	Custom implementation in Python
	Hardware specification	Intel Core i7, 2.10 GHz, 16 GB RAM
	Operating system	Windows 11

Table 2. Value of parameters for FS with and without the proposed algorithm.

	Accuracy	Recall	Precision	F1-score
Without FS	98.33	98.12	98.86	98.62
Proposed algorithm based FS	99.63	99.73	99.57	99.56

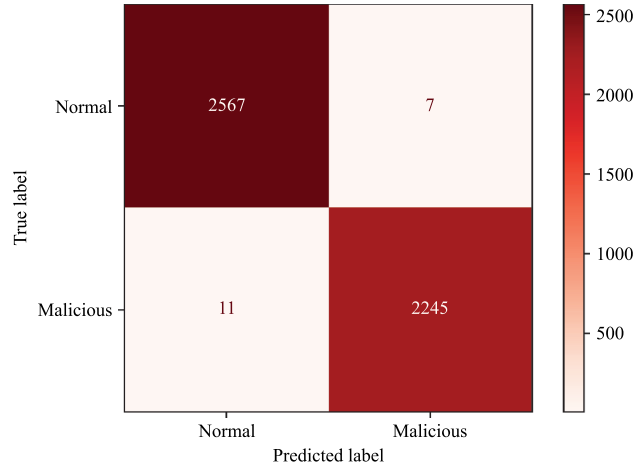
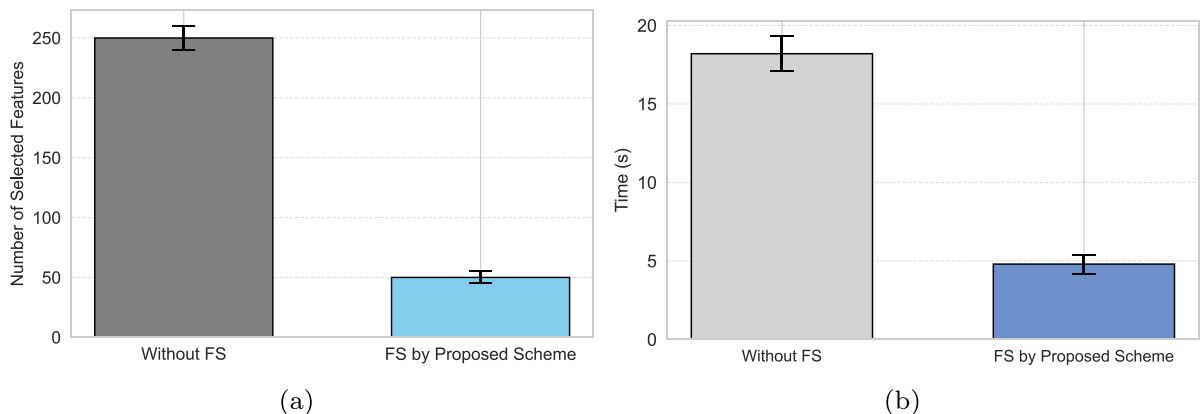
**Figure 4.** Confusion matrix for the proposed algorithm.

Figure 5 shows the significance of the proposed algorithm with respect to reduction in dimensionality and time. Figure 5(a) shows the reduction in the number of features when the proposed algorithm is used. Note that with the reduction in a number of features, the computational complexity of the algorithm is significantly reduced, thus significantly enhancing the efficiency of the IDS system. Figure 5(b) shows the reduction in time for feature selection when the proposed algorithm is used. Note that with the reduction in the number of features, the time used for the same is significantly reduced, further reducing the computational complexity, thereby contributing to the efficiency of the IDS system. Table 2.

4.1. Performance evaluation

Table 4 shows the values of different parameters when the proposed feature selection algorithm is applied and compared to the scenario when no feature selection algorithm is applied. Note that there is significant performance gain by applying the proposed algorithm for feature selection. In Figure 6, the performance of

**Figure 5.** (a) Reduction of dimension using the proposed algorithm, (b) reduction of time using the proposed algorithm.

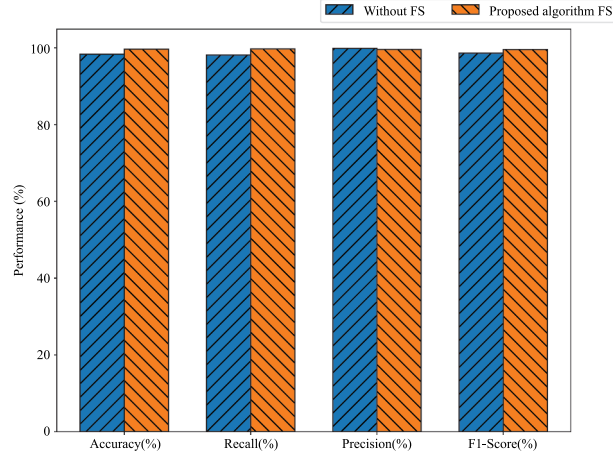


Figure 6. Performance of the IDS based on FS with and without the proposed algorithm.

an IDS in IoT-based WSN is shown when features are selected with and without the proposed algorithm. Note that the proposed approach employs the BH-based method to effectively reduce dimensionality. This improves the efficiency of the system while increasing the accuracy of intrusion detection. The method is particularly effective when applied to the complex AWID3 real-world dataset. The proposed algorithm performs better in terms of accuracy, precision, recall, and F1-score, which are generally accepted and standard parameters to judge an algorithm's performance. The accuracy is calculated using Eq. 4. Precision, recall, and F1-score are calculated using the Eqs. 11, 12, and 13, respectively.

$$\text{Precision} = \frac{T_P}{T_P + F_P} \quad (11)$$

$$\text{Recall} = \frac{T_P}{T_P + F_N} \quad (12)$$

$$F1 - \text{score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (13)$$

Figure 7 shows the comparison of accuracy as calculated using Eq. 4 of the proposed algorithm with the comparison algorithms. It can be observed from the figure that the proposed algorithm has a better accuracy than the other algorithms. In other words, higher accuracy in intrusion detection demonstrates its ability to classify network traffic or system events more reliably than other similar

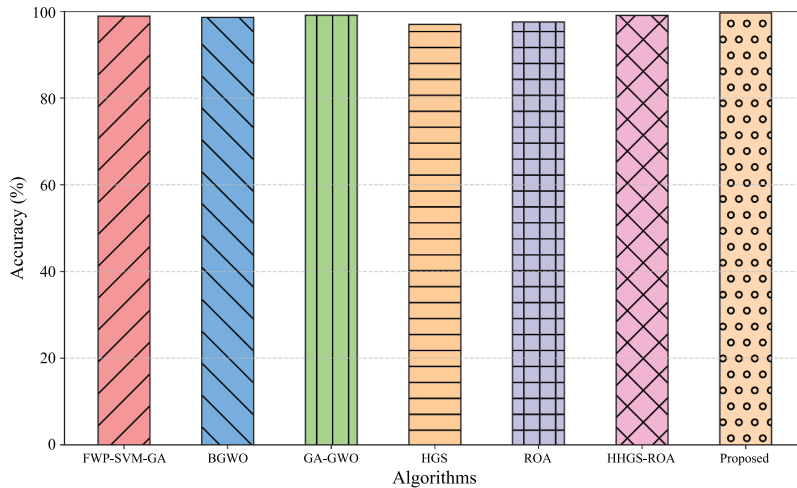


Figure 7. Comparison of proposed algorithm and existing algorithm with respect to accuracy.

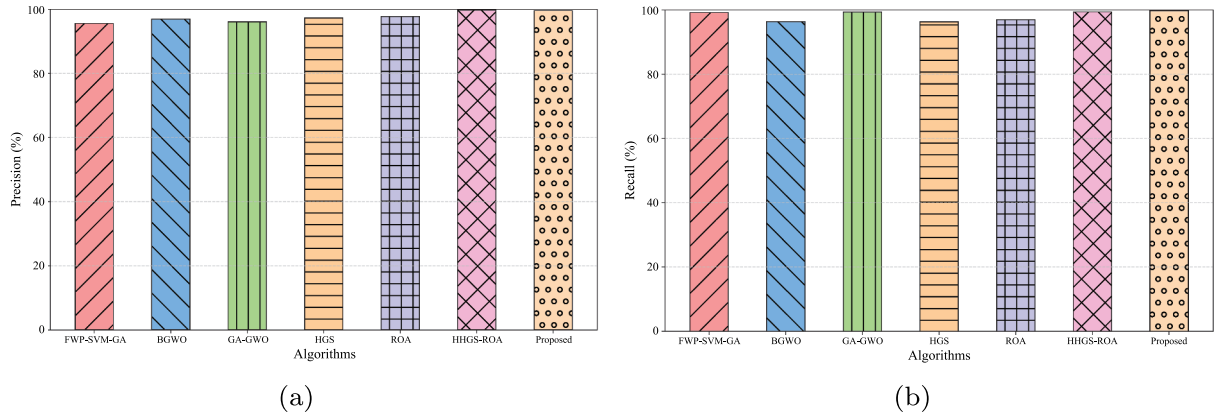


Figure 8. Comparison of proposed algorithm and existing algorithm with respect to (a) precision, (b) Recall.

algorithms. It effectively reduces errors, such as false positives (misclassifying benign activity as malicious) and false negatives (failing to detect actual intrusions). This is attributed to optimal feature selection using efficient star encoding and the novel fitness function used for evaluation. Higher accuracy enhances system security by ensuring that potential threats are detected and mitigated more effectively, minimizing the risk of successful attacks. It reduces the operational burden by generating fewer false alarms, allowing security teams to focus on genuine threats. Additionally, improved accuracy builds trust in the system's reliability, making it more attractive for deployment in critical and sensitive environments.

Figure 8(a) and (b) show the comparison of precision and recall as calculated using Eq.11 and Eq. 12, respectively, of the proposed algorithm with the comparison algorithms. It can be observed from the figure that the proposed algorithm has better precision and recall than the other algorithms. Note that efficient preprocessing of the data and the optimal selection of features using the star encoding is the rationale for better precision and recall for intrusion detection. Improved precision reduces operational noise by cutting down on false alerts, allowing the security system to concentrate on real threats. It enhances system reliability and user trust, as fewer benign actions are incorrectly flagged. Additionally, better precision minimizes wasted resources on unnecessary investigations, leading to more efficient intrusion management. Further, higher recall ensures that all significant threats are detected, leaving minimal room for undetected intrusions. It boosts confidence in the system's coverage, making it suitable for high-risk environments. Moreover, better recall contributes to long-term security by ensuring a proactive response to emerging and sophisticated attacks.

$$FPR = \frac{F_p}{F_p + T_N} \quad (14)$$

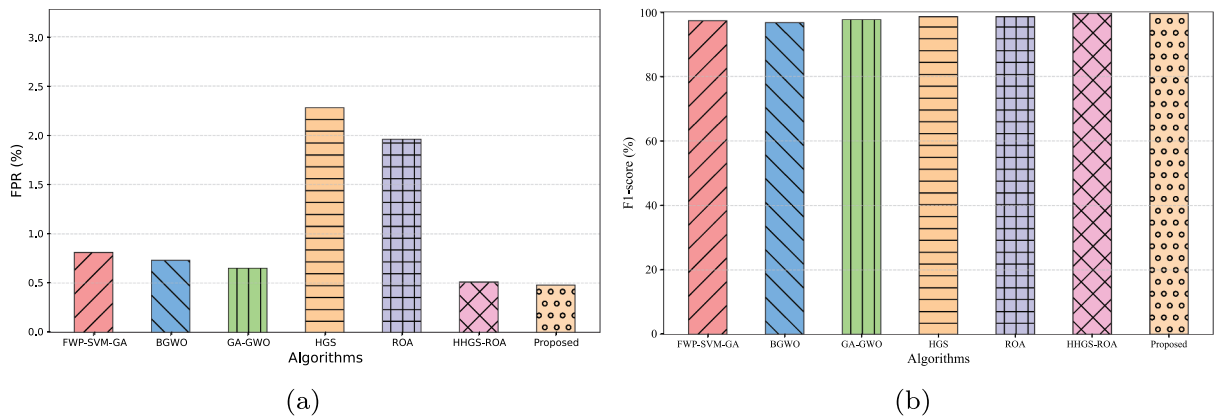


Figure 9. Comparison of proposed algorithm and existing algorithm with respect to (a) FPR, (b) F1-score.

Figure 9(a) and (b) show the comparison of FPR and F1-score as calculated using Eq.14 and Eq. 13, respectively, of the proposed algorithm with the comparison algorithms. It can be observed from the figure that the proposed algorithm has a better FPR and F1-score than the other algorithms. As mentioned earlier, again, the efficient preprocessing of the data set and the optimal selection of features using the BH algorithm contribute to better FPR and F1-score compared to the existing algorithms.

A lower FPR for intrusion detection means the system makes fewer mistakes in flagging benign activity as malicious. Compared to similar algorithms, a lower FPR indicates the algorithm is better at avoiding unnecessary alerts and reducing operational noise. This leads to a more efficient and reliable system where security teams can focus on real threats without being overwhelmed by false alarms. Further, lower FPR reduces the workload on security teams by minimizing false alarms, allowing them to focus on real threats. It enhances user trust in the system by ensuring benign actions are not frequently misclassified as malicious. Additionally, a low FPR saves resources and improves the operational efficiency of the IDS. On the other hand, the F1-score balances precision and recall, making it a comprehensive measure of an algorithm's performance. A higher F1-score means the algorithm effectively minimizes both false positives and false negatives, achieving a balance between sensitivity (recall) and specificity (precision). Compared to other algorithms, a better F1-score highlights its overall effectiveness in correctly detecting intrusions while avoiding unnecessary alerts. Moreover, a higher F1-score ensures that the algorithm provides a robust and balanced IDS. It minimizes the chances of undetected intrusions and false alarms, leading to improved system reliability and protection. Furthermore, better F1-scores foster confidence in deploying the algorithm in critical environments requiring high precision and recall.

Figure 10 compares the False Negative Rate (FNR) across various algorithms, exhibiting that the proposed method achieves the lowest FNR of all. In particular, the proposed method reduces the false negative rate by approximately 56.3% compared to the next-best algorithm (HHGS-ROA). Other algorithms show relatively higher FNR values, with some, such as HGS and ROA, performing significantly worse in this regard. This clear reduction in FNR by the proposed approach can be attributed to its optimized feature selection process, which effectively identifies subtle indicators of intrusions, minimizing instances where real threats are missed. The practical effect of this outcome is enhanced security, as the proposed system is far less likely to overlook actual malicious activities, thereby improving the reliability and trustworthiness of intrusion detection in IoT environments.

Figure 11 compares the proposed algorithm, which stands out with the highest detection rate, as shown by its corresponding distinct pattern and strong border. In particular, the proposed method improves the true positive rate by approximately 0.63% over the next-best algorithm (GA-GWO). Most of the comparative approaches display slightly lower (True Positive Rate) TPRs, highlighting their relative limitations in consistently and accurately identifying genuine intrusions. The superior TPR of the proposed method is largely due to the synergy between the Black Hole Algorithm-based

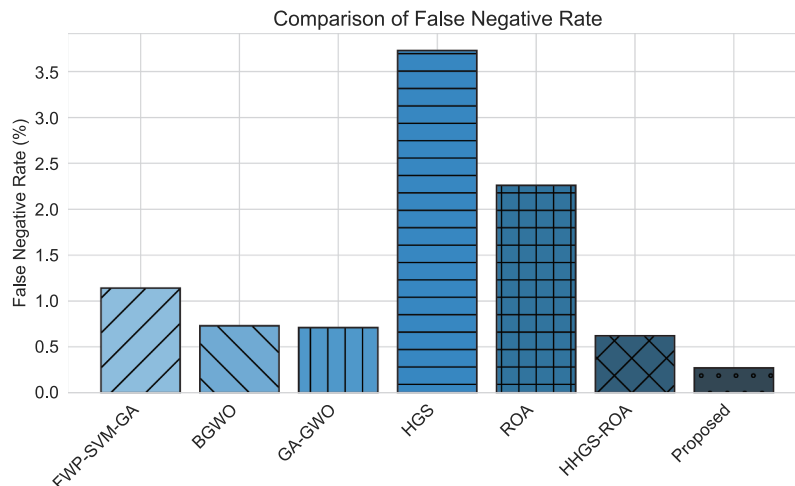


Figure 10. Comparison of proposed algorithm and existing algorithm with respect to FNR.

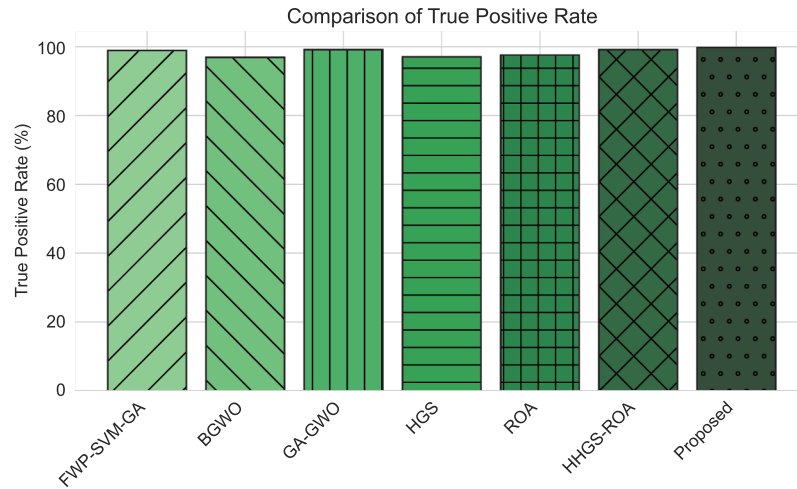


Figure 11. Comparison of proposed algorithm and existing algorithm with respect to TPR.

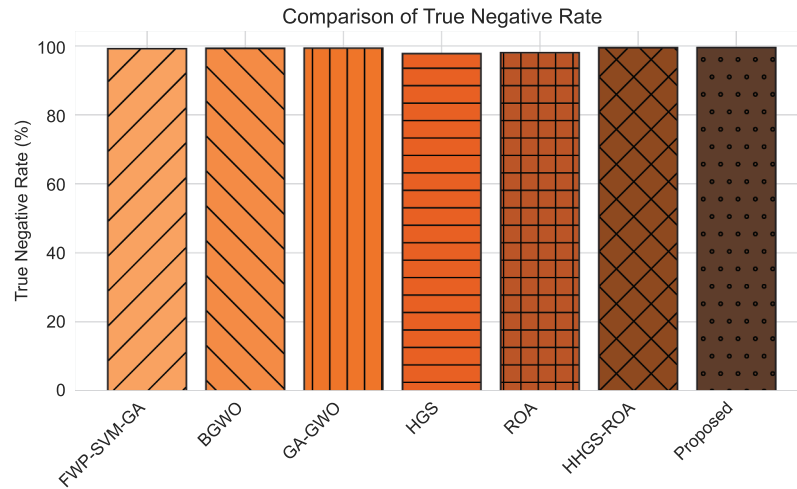


Figure 12. Comparison of proposed algorithm and existing algorithm with respect to TNR.

feature selection and the robust XGBoost classifier, which together enhance the learning of relevant intrusion patterns from complex IoT traffic. Consequently, this leads to a system that not only identifies almost all legitimate threats but also demonstrates a marked improvement over previous state-of-the-art methods in proactive network defense.

Figure 12 compares the True Negative Rate (TNR) bar plot, which reveals that the proposed algorithm achieves the highest rates among all algorithms, evident by its unique pattern and solid black-edged bar. In particular, the proposed method increases the True Negative Rate by approximately 0.02% over the next-best algorithm (HHGS-ROA). While some existing algorithms approach similar TNR values, a few, such as HGS, lag considerably, indicating higher rates of false alarms in those cases. The strong TNR performance of the proposed solution stems from its carefully tailored feature selection, which helps the XGBoost classifier distinguish legitimate network activity from attacks with greater precision. As a result, this translates into fewer false positives, reducing unnecessary alerts and supporting more efficient and accurate network monitoring suitable for real-time, resource-limited IoT deployments.

The proposed algorithm has a better score in terms of parameters than the existing algorithms. The values are listed in Table 3 to better portray the performance of the proposed algorithm.

Table 3. Comparison of the values of different parameters of the proposed algorithm and the comparison algorithms.

Algorithm	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	FPR (%)
FWP-SVM-GA	98.86	95.58	99.18	97.28	0.81
BGWO	98.63	96.89	96.32	96.65	0.73
GA-GWO	99.12	96.07	99.29	97.64	0.65
HGS	97.02	97.25	96.27	98.52	2.28
ROA	97.52	97.74	96.92	98.52	1.96
HHGS-ROA	99.10	99.74	99.38	99.57	0.51
Proposed	99.63	99.57	99.73	99.56	0.49

Table 4. Paired t -test results, effect sizes, and 95% confidence intervals.

Metric	Baseline	Mean Diff (%)	p -value	Cohen's d	95% CI (%)
Accuracy	FWP-SVM-GA	+0.77	< 0.0001	2.15	[0.51, 1.03]
	BGWO	+1.00	< 0.0001	2.98	[0.72, 1.28]
	GA-GWO	+0.51	0.0002	1.96	[0.35, 0.67]
	HGS	+2.61	< 0.0001	8.87	[1.91, 3.33]
	ROA	+2.11	< 0.0001	7.45	[1.45, 2.77]
	HHGS-ROA	+0.53	0.0008	2.08	[0.31, 0.73]
Precision	FWP-SVM-GA	+3.99	< 0.0001	7.10	[2.74, 4.39]
	BGWO	+2.68	< 0.0001	4.53	[1.52, 3.84]
	GA-GWO	+3.50	< 0.0001	6.12	[2.02, 4.78]
	HGS	+2.32	< 0.0001	4.05	[1.12, 3.44]
	ROA	+1.83	0.0003	3.19	[0.62, 2.94]
	HHGS-ROA	−0.17	0.1859	0.62	[−0.39, 0.05]
Recall	FWP-SVM-GA	+0.55	0.0006	1.42	[0.19, 0.91]
	BGWO	+3.41	< 0.0001	13.94	[2.85, 4.15]
	GA-GWO	+0.44	0.0011	1.15	[0.12, 0.82]
	HGS	+3.46	< 0.0001	12.30	[2.90, 3.98]
	ROA	+2.81	< 0.0001	9.15	[2.11, 3.38]
	HHGS-ROA	+0.35	0.0023	1.02	[0.08, 0.62]
F1-score	FWP-SVM-GA	+2.28	< 0.0001	4.38	[1.01, 2.87]
	BGWO	+2.91	< 0.0001	6.52	[1.22, 3.60]
	GA-GWO	+1.92	0.0005	3.75	[0.81, 2.49]
	HGS	+1.04	0.0021	2.14	[0.30, 1.78]
	ROA	+2.04	< 0.0001	4.17	[0.89, 2.69]
	HHGS-ROA	−0.01	0.0874	0.58	[−0.46, 0.15]

4.2. Statistical validation of BHA – IDS performance

The statistical analysis was conducted using 10-fold cross-validation to generate paired performance samples for the proposed Black Hole Algorithm – based IDS (BHA – IDS) versus six baseline methods (FWP – SVM – GA, BGWO, GA – GWO, HGS, ROA, HHGS – ROA) across four metrics (Accuracy, Precision, Recall, F1-score) as shown in Table 4. Paired t -tests at a significance level of $\alpha = 0.05$ were applied to each algorithm pair for each metric, effect sizes were calculated using Cohen's d , and 95% bootstrap confidence intervals for mean differences were estimated to assess the robustness of observed improvements.

Results show that 22 out of 24 paired comparisons (91.7%) yielded $p < 0.05$, confirming statistically significant gains for BHA – IDS in most tests, with Cohen's d values uniformly large or very large ($d > 0.5$, most $d > 0.8$), indicating practical significance. The two non-significant cases (Precision and F1-score vs. HHGS – ROA) still exhibited large effect sizes ($d \approx 0.6$), suggesting comparable performance rather than inferior outcomes. The 95% bootstrap confidence intervals for the mean performance differences further reinforce the consistency and reliability of these findings, with intervals consistently above zero for significant cases. Notably, BHA – IDS demonstrated especially strong gains in Accuracy and Recall, metrics critical for minimizing undetected intrusions and reducing false alarms in IoT security. These outcomes underscore the effectiveness and robustness of BHA – IDS and its superiority over the evaluated baseline methods in multiple aspects of IDS performance.

These findings demonstrate that BHA – IDS delivers genuine and substantial improvements over existing intrusion – detection approaches in IoT environments, as evidenced by both statistical and practical significance across multiple performance dimensions.

5. Discussion

Real-world deployment of the proposed Black Hole Algorithm-based intrusion detection system (IDS) in IoT environments must account for constrained device resources, network variability, and operational overhead; while our framework's efficient feature selection and XGBoost classification minimize computational and memory demands, edge devices with limited RAM and CPU may require model compression (e.g. pruning or quantization) and adaptive scheduling of feature selection to balance detection latency and energy consumption, and low-bandwidth, lossy networks may necessitate hierarchical edge-cloud architectures to ensure timely threat alerts. Additionally, the system must demonstrate robustness across diverse IoT protocols and device manufacturers, as heterogeneous environments can introduce compatibility issues that may compromise detection accuracy or require extensive customization for different deployment scenarios. Furthermore, maintaining consistent performance during network partitions or device failures requires sophisticated fault-tolerance mechanisms and distributed processing capabilities that can significantly increase implementation complexity and operational costs.

Despite the superior performance demonstrated in controlled experimental conditions, the proposed approach faces several inherent limitations that may impact its practical applicability. The evaluation is primarily conducted on the AWID3 dataset, which, while comprehensive, may not fully represent the diverse attack vectors and traffic patterns encountered in real-world IoT deployments across different industrial sectors and geographical regions. The Black Hole Algorithm's population-based optimization approach, although efficient, may still incur computational overhead that could be prohibitive for extremely resource-constrained IoT devices with minimal processing capabilities and memory limitations.

AWID3 is specifically designed for IEEE 802.11 (Wi-Fi) networks, focusing on management- and data-frame attacks unique to Wi-Fi environments, such as deauthentication, disassociation, and KRACK/Kr00k vulnerabilities. As a result, the dataset does not capture traffic characteristics or attack vectors present in other IoT communication protocols (e.g. ZigBee, LoRaWAN, Bluetooth Low Energy), each of which exhibits distinct frame structures, timing behaviors, and security mechanisms. Consequently, while the proposed BHA-IDS demonstrates strong performance on Wi-Fi traffic, its generalizability to non-Wi-Fi IoT settings remains untested and may require protocol-specific feature extraction and model retraining. It would be interesting to evaluate transferability by applying and adapting the framework to datasets from other IoT protocols, incorporating cross-domain generalization strategies such as domain adaptation and multi-protocol feature learning to ensure robust intrusion detection across diverse IoT network technologies.

6. Conclusion

This paper presented a BH algorithm-based efficient feature selection algorithm for an IDS in IoT networks. The scheme had three stages: 1) Data preparation phase, where the initial dataset was preprocessed and cleaned to mitigate the computational overhead for superfluous data. 2) Feature selection using the proposed algorithm, where an optimal set of features of the dataset was obtained. 3) Classification where the selected features led to efficient identification of normal and malicious traffic. The algorithm was applied to the AWID3 dataset to validate its working. Finally, the simulation results of the proposed algorithm and existing algorithms on five different parameters revealed superior performance.

Although the proposed Black Hole Algorithm-based intrusion detection system exhibits strong performance in simulations, its practical deployment and broader applicability are hindered by several limitations. The system's evaluation is confined to the AWID3 dataset, which may not adequately represent the diverse attack vectors, traffic patterns, and protocol variations found in real-world IoT deployments, thus restricting its generalizability and potentially compromising detection accuracy. Additionally, the computational overhead of the population-based optimization process may surpass the capabilities of ultralow-power IoT devices, and the approach does not address adversarial robustness or scalability concerns for real-time operation in large-scale networks. Future work should focus on developing lightweight model variants, comprehensive benchmarking on commercial microcontrollers, and exploration of intelligent hybrid edge-cloud architectures to ensure robust and efficient intrusion detection in resource-constrained IoT environments.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Author contribution

Neeraj Kumar: Conceptualization, Methodology, Original draft preparation, Visualization, Software, Investigation.
 Jyoti Prakash Singh: Conceptualization, Methodology, Original draft preparation, Visualization, Software, Investigation.
 Prabhat Kumar: Conceptualization, Visualization, Methodology, Writing - Review & Editing.

Consent to publish

All authors have read and agreed to the published version of the manuscript.

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